CLASS 8 — SCIENCE TWO-CHAPTER TEST SET C

Chapters 9 & 10: **The Amazing World of Solutes, Solvents and Solutions, Light: Mirrors and Lenses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A liquid of density 2.5 g/cm³ has volume 40 cm³. Its mass is: 1

(a) 16 g **(b)** 102.5 g **(c)** 200 g **(d)** 100 g

2. When two liquids mix, the solute is the one: 1

(a) that floats **(b)** that is coloured
(c) present in smaller amount **(d)** present in larger amount

3. For most solids in liquids, solubility with temperature: 1

(a) stays same **(b)** increases **(c)** decreases **(d)** becomes zero

4. Mass is the amount of: 1

(a) space occupied **(b)** force on object
(c) heat in an object **(d)** matter in an object

5. The curved surface of a liquid in a measuring cylinder is the: 1

(a) bubble **(b)** meniscus **(c)** crest **(d)** trough

6. A ray strikes a mirror at an angle of incidence of 40°. The angle between the reflected ray and the mirror surface is: 1

(a) 80° **(b)** 50° **(c)** 100° **(d)** 40°

7. According to the laws of reflection, angle of incidence is: 1

(a) unrelated **(b)** equal to angle of reflection
(c) half the angle of reflection **(d)** twice the angle of reflection

8. The image in a plane mirror is: 1

(a) inverted and larger **(b)** diminished and real
(c) inverted and smaller **(d)** erect and the same size as the object

9. A concave mirror makes light rays: 1

(a) stay parallel **(b)** diverge **(c)** converge **(d)** disappear

10. Convex mirrors are preferred for traffic observation because they: 1

(a) absorb light **(b)** invert the images
(c) enlarge the images **(d)** give a larger view area

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. State how density changes with temperature. 2

12. Give one example each of a solid–liquid and liquid–liquid solution. 2

13. What is the nature of the image formed by a concave lens? 2

14. Which type of mirror can give both erect and inverted images? Explain briefly. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Describe a solid–solid solution with example. 3

16. State the type of lens for correcting short-sightedness and its action. 3

17. A cube of side 2 cm has mass 24 g. Find its density. 3

OR

Why does a boat made of iron float but an iron nail sinks? 3

18. Is it possible to get the image of an object on a screen with a plane mirror? Why? 3

OR

A light ray incident at 50° falls on a mirror. Find the angle between incident and reflected rays. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Calculate the density of three liquids with the following (mass, volume): A (50 g, 50 mL), B (80 g, 100 mL), C (120 g, 100 mL). Which will float on water? 5

OR

A cold fizzy drink is opened in a hot place. Explain what happens with solubility of gases. 5

20. Describe the use of mirrors by dentists and ENT specialists and why not convex or plane mirrors. 5

OR

Discuss why the rear-view mirror of a bike is convex and what its limitation is. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND